

Homework 3

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Problem 1

Given

$$G = \frac{4\pi\eta}{\int_0^{2\pi} d\varphi \int_0^\pi d\vartheta \sin\vartheta f(\vartheta, \varphi)}$$

$$f(\theta, \varphi) = \begin{cases} 1, & \text{for } \theta < \theta_0 \\ 0, & \text{for } \theta > \theta_0 \end{cases}$$

$$\eta = 1$$

so

$$G = \frac{4\pi}{\int_0^{2\pi} \int_0^{\theta_0} \sin\theta \, d\theta d\varphi}$$

integrating,

$$\int_0^{\theta_0} \sin\theta \, d\theta = [-\cos\theta]_0^{\theta_0} = 1 - \cos\theta_0$$

$$\int_0^{2\pi} d\phi = 2\pi$$

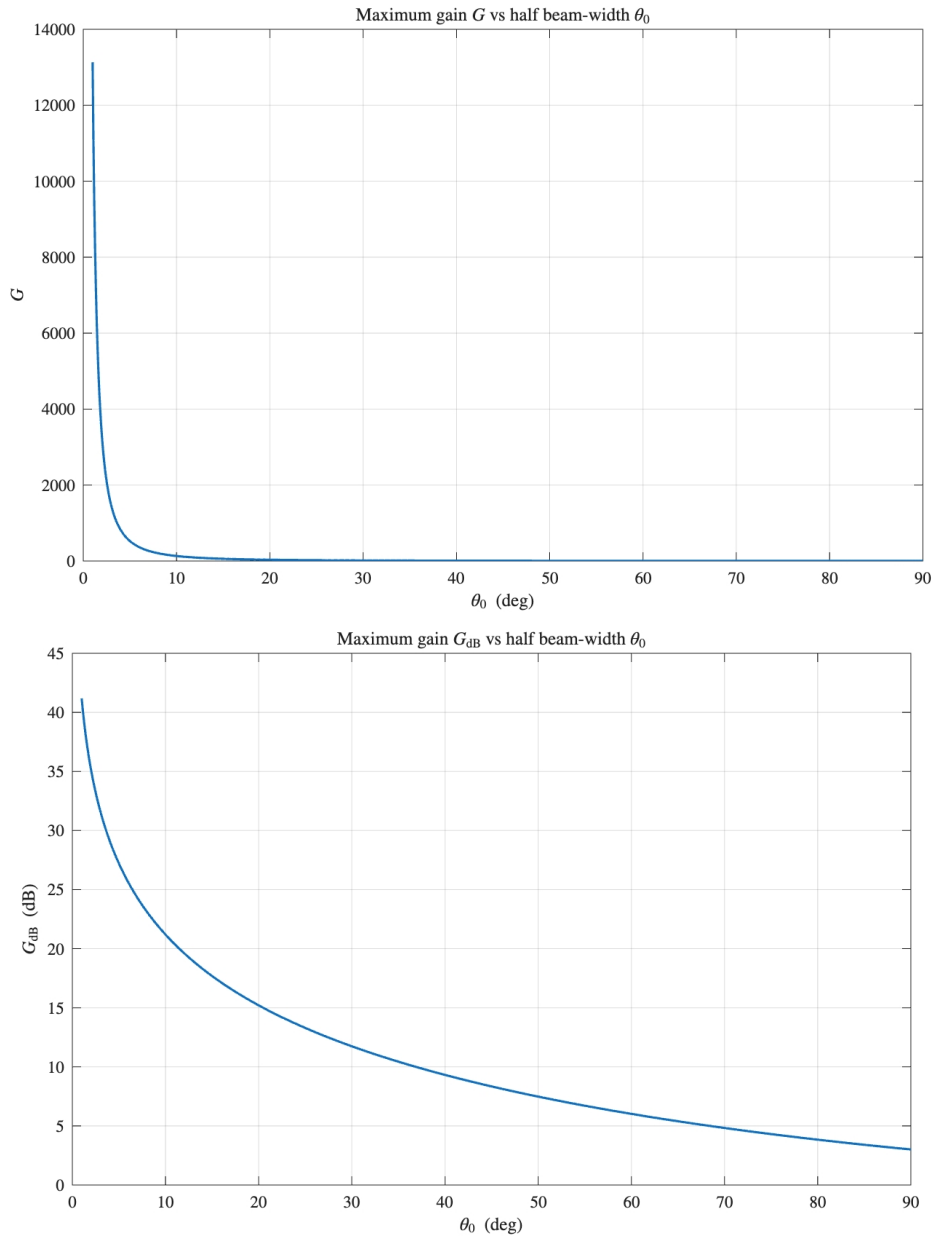
so

$$G = \frac{2}{1 - \cos\theta_0}$$

For plotting we use

$$G_{dB} = 10 \log_{10} G = 10 \log_{10} \left(\frac{2}{1 - \cos\theta_0} \right)$$

Then we get:



The figure shows that the maximum gain decreases when the half beam-width θ_0 increases.

When θ_0 is small, the beam is very narrow, so the radiated power is concentrated in a small angular region and the gain is high. When θ_0 becomes larger, the power is spread over a wider region, so the gain becomes lower. At $\theta_0 = 90^\circ$, the gain is the minimum in this range, about 3 dB.

Problem 2

From the given:

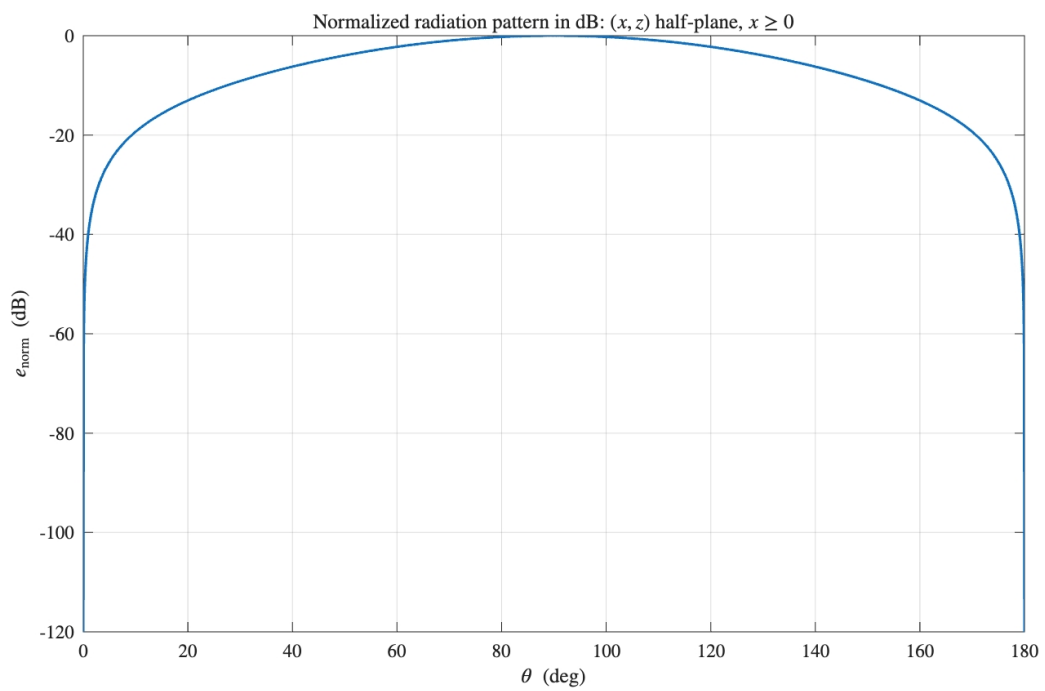
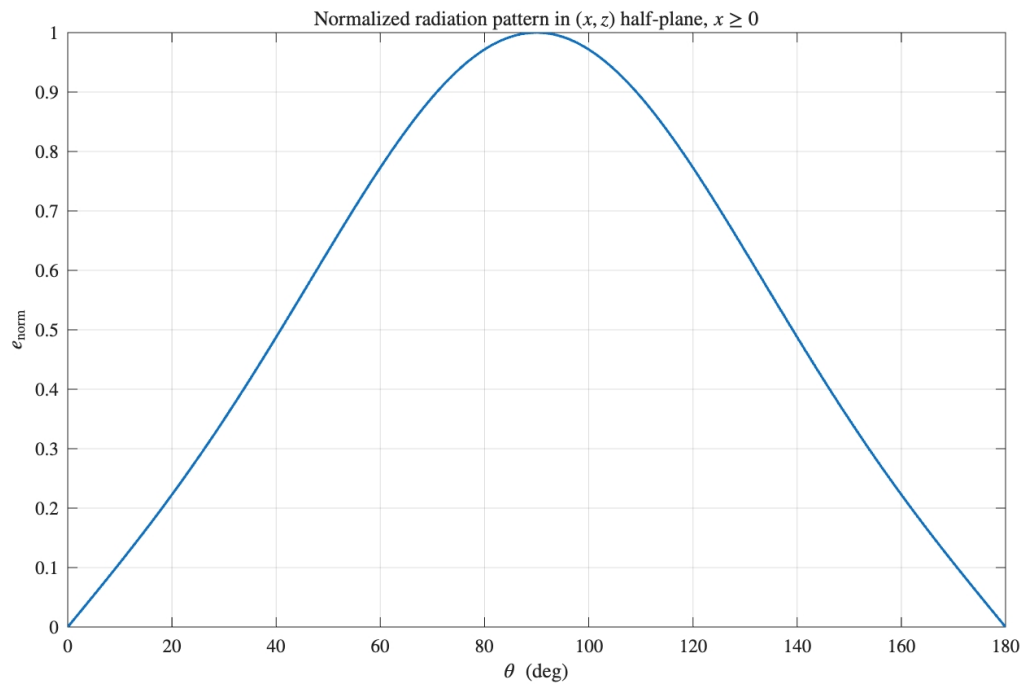
$$e_{\text{norm}}(\theta, \varphi) = \frac{|e(\theta, \varphi)|}{e_{\text{max}}}$$

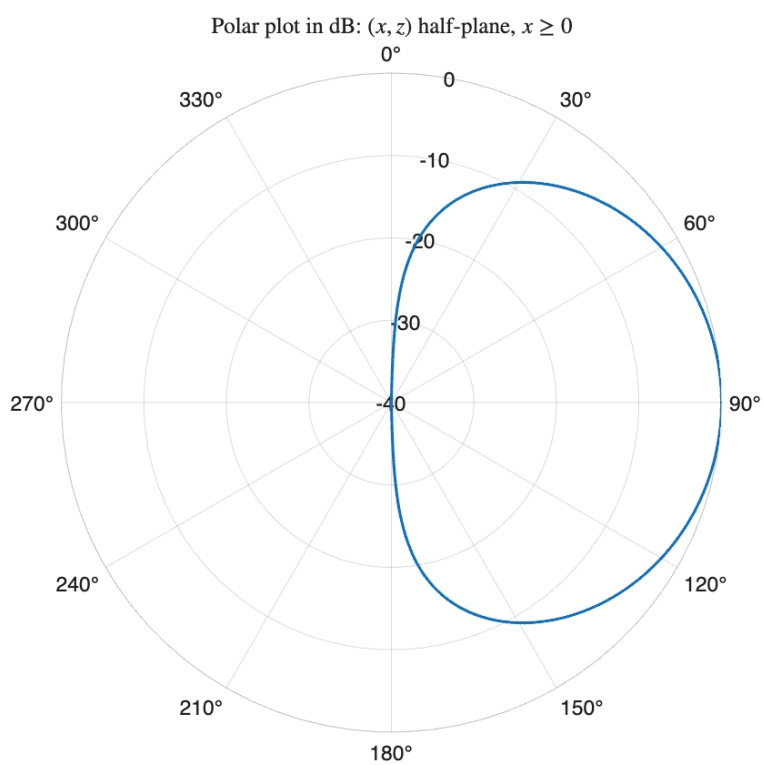
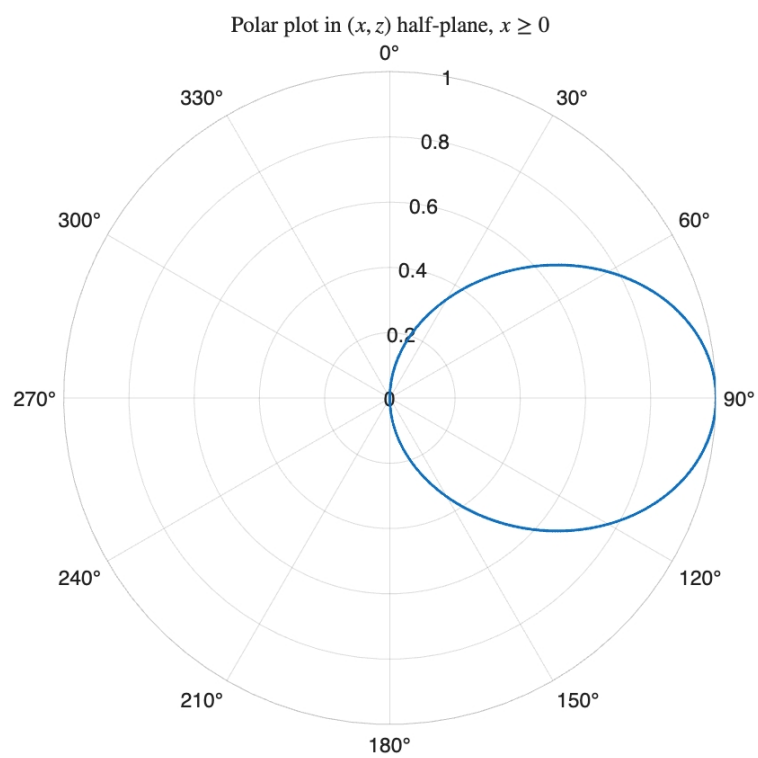
$$e_{\max} = |e(\pi/2, 0)|$$

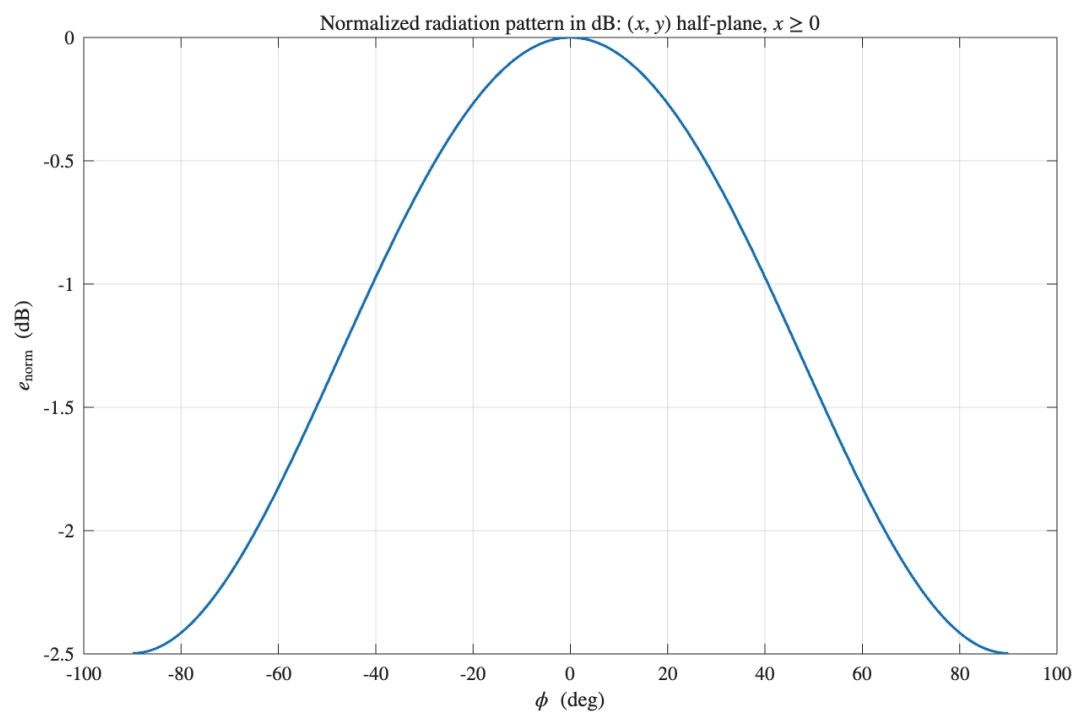
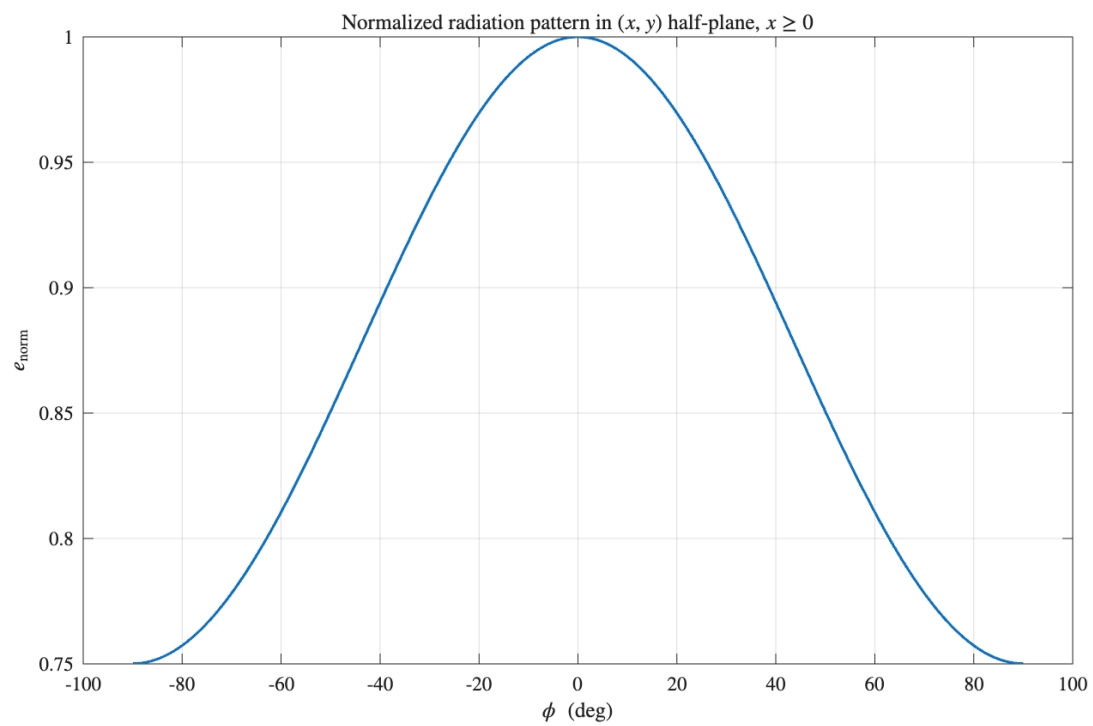
we get:

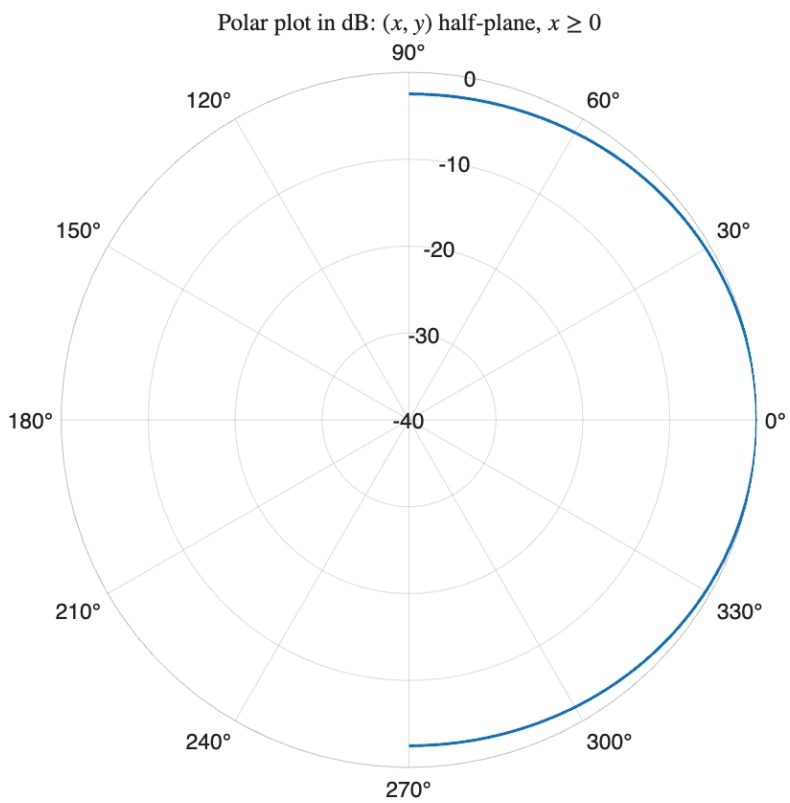
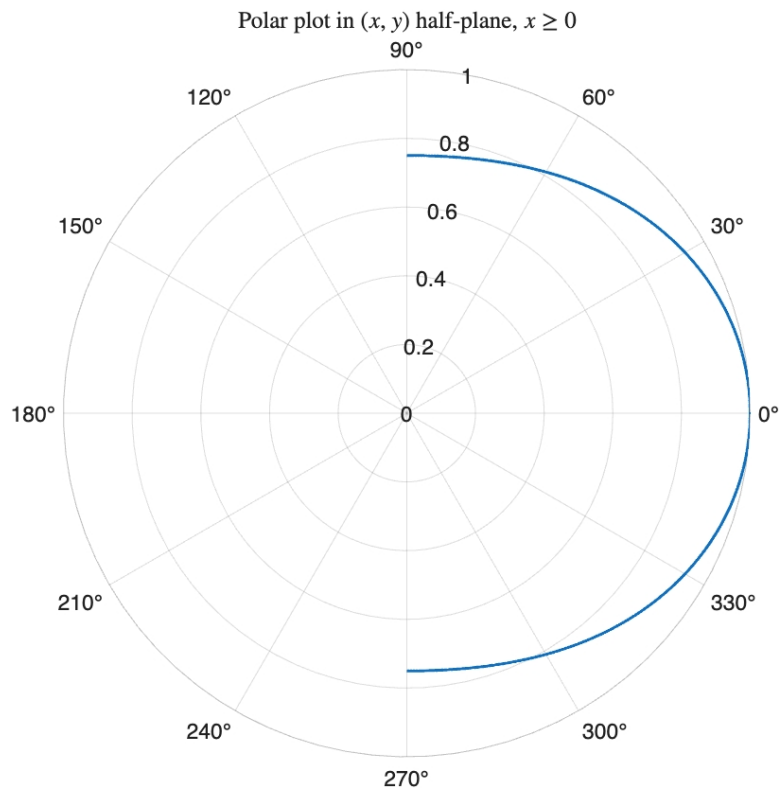
$$e_{\text{norm}}(\theta, \varphi) = \left| \sin \theta \operatorname{sinc} \left(\pi \frac{W}{\lambda_0} \cos \theta \right) \cos \left(\pi \frac{L}{\lambda_0} \sin \theta \sin \varphi \right) \right|$$

Then we use matlab to plot the figures:









From the given radiation vector, the maximum radiation is obtained for

$$(\theta, \varphi) = \left(\frac{\pi}{2}, 0 \right)$$

that is along the +x direction.

Since the radiated field is polarized along $\hat{\varphi}$, and $\hat{\varphi} = -\hat{x} \sin \varphi + \hat{y} \cos \varphi$, at $\varphi = 0$ we have $\hat{\varphi}|_{\varphi=0} = \hat{y}$. Therefore, in the direction of maximum radiation, the electric field is directed along +y.

Hence, the E-plane is the **(x,y) plane**:

Using the far-field relation:

$$\hat{H} \propto \hat{r} \times \hat{E}$$

and considering that the maximum radiation direction is along $\hat{x}$ while the electric field is along $\hat{y}$, it follows that

$$\hat{x} \times \hat{y} = \hat{z}$$

so the magnetic field is directed along +z.

Hence, the H-plane is the **(x,z) plane**.

Problem 3

a)

From our class we know:

$$P_{rad} = \frac{1}{2} \frac{1}{Z_0} \frac{1}{(4\pi r)^2} \int_0^{2\pi} d\varphi \int_0^\pi r^2 \sin \theta |e(\theta, \varphi)|^2 d\theta$$

as is given in the problem:

$$|e(\theta, \varphi)|^2 = 4Z_0^2 |I_0|^2 \sin^3 \theta$$

substituting:

$$P_{rad} = \frac{Z_0 |I_0|^2}{8\pi^2} \int_0^{2\pi} d\varphi \int_0^\pi \sin^4 \theta d\theta$$

we do the integration:

$$\int_0^{2\pi} d\varphi = 2\pi, \quad \int_0^\pi \sin^4 \theta d\theta = \frac{3\pi}{8}$$

$$P_{rad} = \frac{Z_0 |I_0|^2}{8\pi^2} \cdot 2\pi \cdot \frac{3\pi}{8}$$

Thus we get

$$P_{rad} = \frac{3}{32} Z_0 |I_0|^2 = \frac{3}{32} \cdot 377 |I_0|^2 \approx 35.34 |I_0|^2$$

b)

As is given in the slides

$$P_{rad} = \frac{1}{2} R_{rad} |I_{in}|^2$$

So

$$R_{rad} = \frac{3}{16} Z_0 = \frac{3}{16} \cdot 377 \Omega \approx 70.7 \Omega$$

Problem 4

From the slides:

$$d(\vartheta, \varphi) = \frac{S(\vartheta, \varphi)}{S_{average}}$$

$$S_{average} = \frac{P_{rad}}{4\pi r^2}$$

thus

$$S(\vartheta, \varphi) = d(\vartheta, \varphi) \frac{P_{rad}}{4\pi r^2}$$

In the direction of maximum radiation, the directivity function reaches its maximum value, which is the gain G

$$d(\vartheta_{max}, \varphi_{max}) = G$$

In this problem

$$P_{rad} = P_{in}$$

substituting

$$S(\vartheta_{\max}, \varphi_{\max}) = G \frac{P_{in}}{4\pi r^2}$$

with

$$G = 10^{38/10} = 6309.57$$

$$P_{in} = 12 \text{ W}, \quad r = 10^4 \text{ m}$$

thus

$$S(\vartheta_{\max}, \varphi_{\max}) = 6309.57 \cdot \frac{12}{4\pi (10^4)^2} \approx 6.03 \times 10^{-5} \text{ W/m}^2$$

In the far-field region

$$S = \frac{|E|^2}{2Z_0}$$

with

$$Z_0 = 120\pi \approx 377 \text{ } \Omega$$

Finally we get

$$|E| \approx 0.213 \text{ V/m}$$

Problem 5

From the slides

$$A_{eq} = G \frac{\lambda^2}{4\pi}$$

with

$$G = 10^{25/10} = 316.23$$

$$\lambda = \frac{c}{f_0} = \frac{3 \times 10^8}{10 \times 10^9} = 0.03 \text{ m}$$

thus

$$A_{eq} \approx 0.02265 \text{ m}^2$$

In addition

$$\left(\frac{dP}{d\Sigma}\right)_{inc} = S = \frac{|E_{inc}|^2}{2Z_0}$$

with

$$|E_{inc}| = 2 \text{ mV/m} = 2 \times 10^{-3} \text{ V/m}$$

$$Z_0 = 120\pi \approx 377 \text{ } \Omega$$

thus

$$\left(\frac{dP}{d\Sigma}\right)_{inc} \approx 5.31 \times 10^{-9} \text{ W/m}^2$$

As is given in the slides

$$P_{av} = \left(\frac{dP}{d\Sigma}\right)_{inc} \cdot A_{eq}$$

thus

$$P_{av} \approx 1.20 \times 10^{-10} \text{ W}$$

For a matched load

$$P_{av} = \frac{|V_0|^2}{8 \text{ Re}\{Z_{RX}\}}$$

with

$$Z_{RX} = (75 - j25) \text{ } \Omega$$

Finally we get

$$|V_0| = \sqrt{8 \times 75 \times 1.20 \times 10^{-10}} \approx 0.268 \text{ mV}$$